

Figure 1: Area plot

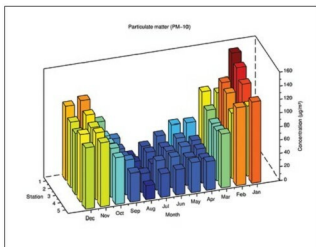


Figure 2: Bar plot 3D

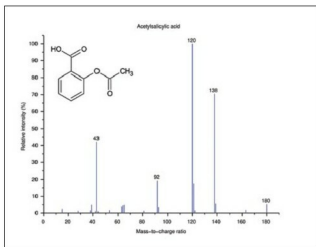


Figure 3: Impulse plot

x axis and y axis labels to the respective axis orientation, since `x_label.font_angle` and `y_label.font_angle` are not specified. The code is partially based on the Scilab code

written by Alex Carneiro (see <http://usingscilab.blogspot.it/2012/09/plot3d2-creating-cube.html>).

Impulse plot

This plot type can be used to plot a mass spectrum. In this example, the acetylsalicylic acid (2-acetoxybenzoic acid) mass spectrum is considered. The peak data are taken from http://sdb.sdb.aist.go.jp/sdb/cgi-bin/direct_frame_top.cgi and then organised in four columns. The first two columns are the *spectrum* coordinates and the second two are the *fragments* coordinates, on which the numerical labels are put. With the LaTeX command `includegraphics` and the primitive `xstring`, it's also possible to add an image of the molecular structure (via the parameter 'g'). In `xstring(5,65,"$"+g+"$")`, first the coordinates are written about the position of 'g' on the plot and then 'g' is inserted between two '\$'. That is how LaTeX can be used with Scilab. If the impulses are plotted horizontally, the result can be adapted to something similar to a Gantt chart.

Radar plot

The radar or spider plot can be used to represent one or more data series in a form similar to a polar plot. A practical example is given by the radar plot widely used by Active House (<http://www.activehouse.info>) for displaying nine parameters about *comfort*, *energy* and the *environment* in the design of a building and in the finished building. Basically, the structure of this plot is built on a bi-dimensional grid via the function `meshgrid`. There are also a lot of settings with the functions `gca`, `gce` and with the primitive `get`. The legend is divided into two parts: the first is written with the primitive `xstring` and the second with the function `legends`. Due to the large legend, the plot has shifted a bit on the left side. The code is adapted to Scilab from the MATLAB code written by Cheng Li (see <http://www.mathworks.com/matlabcentral/fileexchange/30305-spider-rader-chart-for-multi-data>).

Ribbon plot

Now let's consider one absorbance spectrum measured from 350 nm to 550 nm (steps of 10 nm each) and at eight different times. So the data table is composed of eight columns. The most important thing about this plot type is how to use the function `cat` within the function `surf`. The result is that each 'z' value (each column) is duplicated because, in this case, it must be a real matrix defining the surface height.

Ternary plot

A ternary plot is used in physical chemistry, petrology, mineralogy, metallurgy and other physical sciences to show the compositions of systems composed of three species. It's also used in population genetics and in game theory [see http://en.wikipedia.org/wiki/Ternary_plot]. In this example, the compositions of certain building materials are shown as in a similar example in the book *Inorganic Industrial Chemistry* by Büchner, Schliebs, Winter, Büchel (see 'References' at the